

Lecture 7: Theories of unconstrained optimization (part 2/2)

MA3236 Non-Linear Programming

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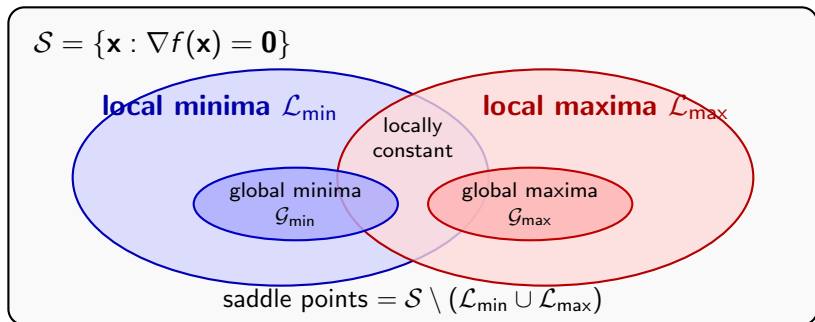
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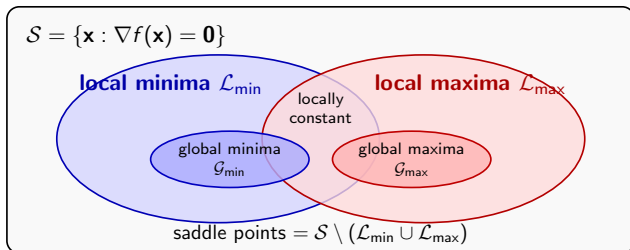
Quick review

Let $f \in C^2(\mathcal{D})$, where $\mathcal{D} \subseteq \mathbf{R}^n$ is open. Lecture 6 established the necessary filters

$$\text{local minimum} \implies \nabla f(\mathbf{x}^*) = \mathbf{0}, \quad \mathbf{H}_f(\mathbf{x}^*) \succeq 0.$$



(Uncompleted) workflow



- 1 Identify the problem at hand is **unconstrained**.
- 2 (Establish that **a global minimizer exists**.)
- 3 Solve $\nabla f(\mathbf{x}) = \mathbf{0}$ for all stationary points.
- 4 If the stationary set is **finite**, compare objective values (to decide global optimality).
- 5 Classify candidates with $\mathbf{H}_f(\mathbf{x})$ **when possible**.

Second-order sufficient condition: one-dimensional cases

At a stationary point x^* ,

$$\phi(x^* + h) - \phi(x^*) = \frac{1}{2}\phi''(x^*)h^2 + o(h^2).$$

If $\phi''(x^*) > 0$, then for all sufficiently small $h \neq 0$, the remainder is smaller than a fixed fraction of $\frac{1}{2}\phi''(x^*)h^2$. Hence the positive quadratic term decides the sign.

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Theorem 7.1 (One-dimensional second-order sufficient condition)

Let ϕ be twice continuously differentiable near an interior point x^ .*

- *If $\phi'(x^*) = 0$ and $\phi''(x^*) > 0$, then x^* is a strict local minimizer.*
- *If $\phi'(x^*) = 0$ and $\phi''(x^*) < 0$, then x^* is a strict local maximizer.*

Second-order sufficient condition: multi-dimensional cases

$\mathbf{A} \succeq 0$: PSD; $\mathbf{A} \succ 0$: PD; $\mathbf{A} \preceq 0$: NSD; $\mathbf{A} \prec 0$: ND.

Theorem 7.2 (Second-order sufficient condition)

Let $\mathcal{D} \subseteq \mathbf{R}^n$ be open, let $f \in C^2(\mathcal{D})$, and let $\mathbf{x}^* \in \mathcal{D}$.

- If $\nabla f(\mathbf{x}^*) = \mathbf{0}$ and $\mathbf{H}_f(\mathbf{x}^*) \succ 0$, then $\mathbf{x}^*$ is a strict local minimizer.
- If $\nabla f(\mathbf{x}^*) = \mathbf{0}$ and $\mathbf{H}_f(\mathbf{x}^*) \prec 0$, then $\mathbf{x}^*$ is a strict local maximizer.

Proof sketch.

$$f(\mathbf{x}^* + \mathbf{d}) - f(\mathbf{x}^*) = \frac{1}{2} \mathbf{d}^T \mathbf{H}_f(\mathbf{x}^*) \mathbf{d} + o(\|\mathbf{d}\|^2), \quad \mathbf{H}_f(\mathbf{x}^*) = \mathbf{Q} \mathbf{\Lambda} \mathbf{Q}^T.$$

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$$= \sum_{i=1}^n \lambda_i \tilde{d}_i^2 \geq \sum_{i=1}^n \left(\min_{1 \leq j \leq n} \lambda_j \right) \tilde{d}_i^2$$

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$$= \left(\min_{1 \leq j \leq n} \lambda_j \right) \|\tilde{\mathbf{d}}\|^2 = \left(\min_{1 \leq j \leq n} \lambda_j \right) \|\mathbf{d}\|^2.$$

The theorem gives local quadratic growth

The proof yields constants $c > 0$ and $r > 0$ such that

$$f(\mathbf{x}) \geq f(\mathbf{x}^*) + c\|\mathbf{x} - \mathbf{x}^*\|^2 \quad \text{whenever } \|\mathbf{x} - \mathbf{x}^*\| < r.$$

Consequently,

- $\mathbf{x}^*$ is a strict local minimizer;
- $\mathbf{x}^*$ is the unique minimizer of f over $B(\mathbf{x}^*, r)$; and
- **objective growth is at least quadratic near $\mathbf{x}^*$.**

An indefinite Hessian identifies a saddle

At a stationary point, suppose $\mathbf{H}_f(\mathbf{x}^*)$ is indefinite. Then there are directions $\mathbf{d}_+$ and $\mathbf{d}_-$ with

$$\mathbf{d}_+^T \mathbf{H}_f(\mathbf{x}^*) \mathbf{d}_+ > 0, \quad \mathbf{d}_-^T \mathbf{H}_f(\mathbf{x}^*) \mathbf{d}_- < 0.$$

Set $\phi_{\mathbf{d}}(t) = f(\mathbf{x}^* + t\mathbf{d})$. Then

$$\phi_{\mathbf{d}}(t) - \phi_{\mathbf{d}}(0) = \frac{t^2}{2} \mathbf{d}^T \mathbf{H}_f(\mathbf{x}^*) \mathbf{d} + o(t^2).$$

Thus nearby values are above $f(\mathbf{x}^*)$ along $\mathbf{d}_+$ and below it along $\mathbf{d}_-$.

Corollary 7.3 (Indefinite Hessian test)

A stationary point with an indefinite Hessian is a saddle point.

The certificate is local, not global

Both functions satisfy

$$\phi_i'(0) = 0, \quad \phi_i''(0) = 2 > 0,$$

so 0 is a strict local minimizer in each case.

No global minimizer

$$\phi_1(x) = x^2 + x^3.$$

Since $\phi_1(x) \rightarrow -\infty$ as $x \rightarrow -\infty$,
no global minimizer exists.

A global minimizer exists

$$\phi_2(x) = x^2 - 3x^4 + x^6.$$

It is coercive, but
 $\phi_2(1) = -1 < 0 = \phi_2(0)$; hence its
global minimizer is not 0.

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Well-posedness + convexity bridges local to global.

Stationarity cannot be omitted

Positive curvature alone is not enough. Let

$$\phi(x) = x + x^2.$$

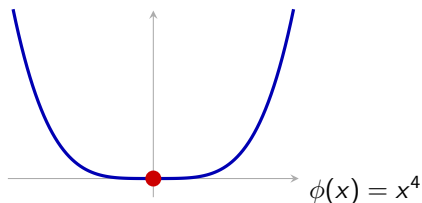
At $x = 0$,

$$\phi''(0) = 2 > 0, \quad \phi'(0) = 1 \neq 0.$$

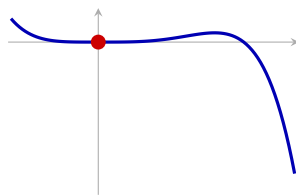
The point 0 is not a local minimizer; the unique global minimizer is $x = -\frac{1}{2}$.

Sufficient, but not necessary

Both functions satisfy $\phi'(0) = \phi''(0) = 0$.



Role of 0: strict global minimum



$$\phi(x) = -x^4 - x^5$$

Role of 0: strict local minimum

Sufficient, but not necessary (cont'd)

Consider the coupled function

$$g(x_1, x_2) = (x_1 + x_2)^2 + x_2^4.$$

At the origin,

$$\nabla g(\mathbf{0}) = \mathbf{0}, \quad \mathbf{H}_g(\mathbf{0}) = 2 \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \succeq 0 \quad \text{is singular.}$$

Since $g(\mathbf{x}) \geq 0$, with equality only at $\mathbf{0}$, the origin is the unique global minimizer despite the singular Hessian.

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Along the kernel direction $\mathbf{d} = (1, -1)^T$,

$$\mathbf{H}_g(\mathbf{0})\mathbf{d} = \mathbf{0}, \quad g(t, -t) = t^4 > 0 \quad (t \neq 0).$$

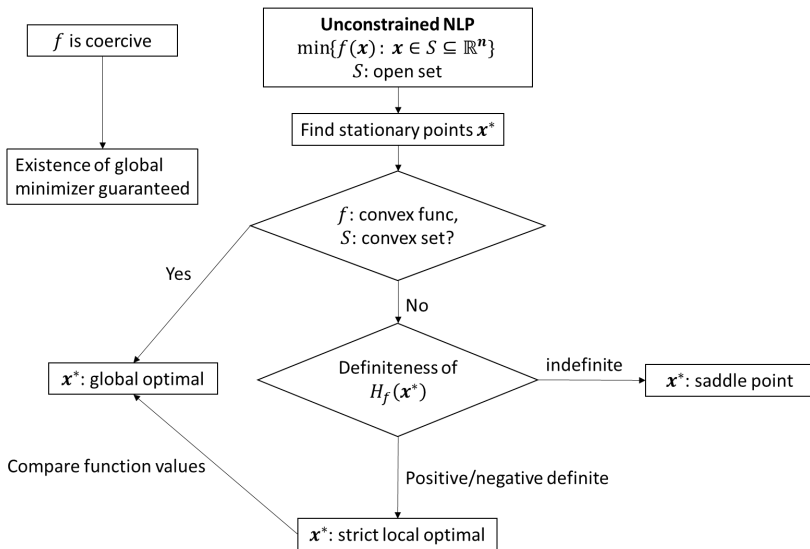
It is strict but has no quadratic growth along this line.

The second-order decision chart is conditional

At an interior stationary point $\mathbf{x}^*$:

Hessian at $\mathbf{x}^*$	conclusion
$\mathbf{H}_f(\mathbf{x}^*) \succ 0$	strict local minimum
$\mathbf{H}_f(\mathbf{x}^*) \prec 0$	strict local maximum
$\mathbf{H}_f(\mathbf{x}^*)$ indefinite	saddle point
singular PSD or NSD	inconclusive

The chart classifies a candidate; it does not find the candidate or prove that a global optimizer exists.



If the Hessian is singular semidefinite, inspect higher-order terms or suitable curves through the point.

Example: a coupled cubic

Example 7.4

Consider

$$F(x_1, x_2) = 2x_1^3 + 6x_1x_2^2 - 6x_1^2 - 6x_2^2 + 2.$$

Find every stationary point and apply the Hessian test.

$$\nabla F(x_1, x_2) = \begin{bmatrix} 6(x_1^2 + x_2^2 - 2x_1) \\ 12x_2(x_1 - 1) \end{bmatrix}.$$

Stationarity is equivalent to

$$x_1^2 + x_2^2 - 2x_1 = 0, \quad x_2(x_1 - 1) = 0.$$

Example: a coupled cubic (cont'd)

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Branch 1: $x_2 = 0$

$$x_1(x_1 - 2) = 0,$$

giving $(0, 0)$ and $(2, 0)$.

Branch 2: $x_1 = 1$

$$x_2^2 = 1,$$

giving $(1, 1)$ and $(1, -1)$.

Hence

$$\mathcal{S} = \{(0, 0), (2, 0), (1, 1), (1, -1)\}.$$

Example: a coupled cubic (cont'd)

$$\mathcal{S} = \{(0, 0), (2, 0), (1, 1), (1, -1)\}.$$

$$\text{From } \nabla F(x_1, x_2) = \begin{bmatrix} 6(x_1^2 + x_2^2 - 2x_1) \\ 12x_2(x_1 - 1) \end{bmatrix},$$

$$\mathbf{H}_F(x_1, x_2) = 12 \begin{bmatrix} x_1 - 1 & x_2 \\ x_2 & x_1 - 1 \end{bmatrix}.$$

At the two stationary points on $x_2 = 0$,

$$\mathbf{H}_F(0, 0) = -12I_2 \prec 0, \quad \mathbf{H}_F(2, 0) = 12I_2 \succ 0.$$

Therefore

- $(0, 0)$ is a strict local maximizer;
- $(2, 0)$ is a strict local minimizer.

Example: a coupled cubic (cont'd)

$$\mathcal{S} = \{(0, 0), (2, 0), (1, 1), (1, -1)\}.$$

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Write $\sigma \in \{1, -1\}$. At $(1, \sigma)$,

$$\mathbf{H}_F(1, \sigma) = 12 \begin{bmatrix} 0 & \sigma \\ \sigma & 0 \end{bmatrix}.$$

Its eigenvalues are

$$12 \quad \text{and} \quad -12.$$

Thus the Hessian is indefinite at both $(1, 1)$ and $(1, -1)$. They are both saddle points.

Example: singular Hessian

Example 7.5

Consider

$$f(x_1, x_2) = (x_1 + x_2^2)^2 - x_2^6.$$

Verify that the origin is stationary and classify it.

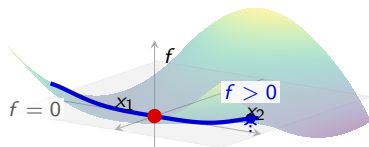
$$\nabla f(\mathbf{x}) = \begin{bmatrix} 2(x_1 + x_2^2) \\ 4(x_1 + x_2^2)x_2 - 6x_2^5 \end{bmatrix}, \quad \mathbf{H}_f(\mathbf{x}) = \begin{bmatrix} 2 & 4x_2 \\ 4x_2 & 4x_1 + 12x_2^2 - 30x_2^4 \end{bmatrix}.$$

Therefore, the origin is stationary and

$$\mathbf{H}_f(\mathbf{0}) = \begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix} \succeq 0, \quad \text{Null}(\mathbf{H}_f(\mathbf{0})) = \text{span}\{(0, 1)^T\}.$$

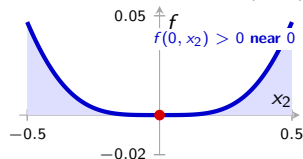
Example: singular Hessian (cont'd)

$$f(x_1, x_2) = (x_1 + x_2^2)^2 - x_2^6, \quad \nabla f(\mathbf{0}) = \mathbf{0},$$
$$\mathbf{H}_f(\mathbf{0}) = \begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix} \succeq 0, \quad \text{Null}(\mathbf{H}_f(\mathbf{0})) = \text{span}\{(0, 1)^T\}.$$



trace over the kernel line: $x_1 = 0$ and $f(0, t) > 0$
for $0 < |t| < 1$

Zoomed projection onto (x_2, f)



Blue trace stays above $f = 0$ for
 $0 < |x_2| \leq 0.5$.

Could a nearby *curved* path decrease f ?

Example: singular Hessian (cont'd)

$$f(x_1, x_2) = (x_1 + x_2^2)^2 - x_2^6.$$

For fixed $x_2 = t$, minimizing the square term over x_1 gives

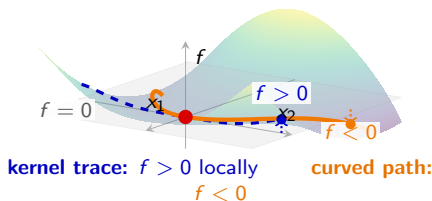
$$x_1^*(t) = -t^2, \quad f(-t^2, t) = -t^6 < 0 \quad (t \neq 0).$$

Example: singular Hessian (cont'd)

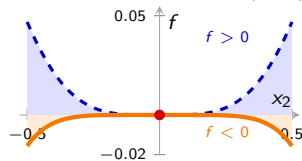
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Zoomed projection onto (x_2, f)



Same x_2 ; opposite signs.

Therefore, **the origin is a saddle point!**

Quadratics are Taylor models with no remainder

Let

$$q(\mathbf{x}) = \frac{1}{2}\mathbf{x}^T \mathbf{Q} \mathbf{x} + \mathbf{c}^T \mathbf{x} + \gamma, \quad \mathbf{Q} = \mathbf{Q}^T.$$

For every displacement $\mathbf{d}$,

$$q(\mathbf{x} + \mathbf{d}) - q(\mathbf{x}) = (\mathbf{Q}\mathbf{x} + \mathbf{c})^T \mathbf{d} + \frac{1}{2}\mathbf{d}^T \mathbf{Q} \mathbf{d}.$$

Thus

$$\nabla q(\mathbf{x}) = \mathbf{Q}\mathbf{x} + \mathbf{c}, \quad \mathbf{H}_q(\mathbf{x}) = \mathbf{Q},$$

and there are no hidden higher-order terms.

Three cases for an unconstrained quadratic

Theorem 7.6 (Complete quadratic minimization classification)

For symmetric $\mathbf{Q}$ and $q(\mathbf{x}) = \frac{1}{2}\mathbf{x}^T \mathbf{Q} \mathbf{x} + \mathbf{c}^T \mathbf{x} + \gamma$ on $\mathbf{R}^n$:

- ① if $\mathbf{Q} \not\geq 0$, then $\inf q = -\infty$;
- ② if $\mathbf{Q} \succeq 0$ and $\mathbf{c} \notin \text{Range}(\mathbf{Q})$, then $\inf q = -\infty$;
- ③ if $\mathbf{Q} \succeq 0$ and $\mathbf{c} \in \text{Range}(\mathbf{Q})$, then

$$\min q = \gamma - \frac{1}{2}\mathbf{w}^T \mathbf{c}, \quad \arg \min q = -\mathbf{w} + \text{Null}(\mathbf{Q}),$$

where $\mathbf{w} \in \mathbf{R}^n$ solves $\mathbf{c} = \mathbf{Q}\mathbf{w}$.

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where $\mathbf{w} \in \mathbf{R}^n$ solves $\mathbf{c} = \mathbf{Q}\mathbf{w}$. If further $\mathbf{Q} \succ 0$, then

$$\min q = \gamma - \frac{1}{2}\mathbf{c}^T \mathbf{Q}^{-1} \mathbf{c}, \quad \arg \min q = \{-\mathbf{Q}^{-1} \mathbf{c}\}.$$

$$\mathbf{Q} \not\preceq 0 \Rightarrow \inf q = -\infty$$

If $\mathbf{Q} \not\preceq 0$, choose $\mathbf{d}$ with

$$\mathbf{d}^T \mathbf{Q} \mathbf{d} < 0.$$

Along the line $\mathbf{x} = t\mathbf{d}$,

$$q(t\mathbf{d}) = \frac{1}{2}t^2(\mathbf{d}^T \mathbf{Q} \mathbf{d}) + t\mathbf{c}^T \mathbf{d} + \gamma.$$

The negative quadratic term dominates the linear term as $|t| \rightarrow \infty$.
Therefore $q(t\mathbf{d}) \rightarrow -\infty$.

$$\mathbf{Q} \succeq 0 \Rightarrow ?$$

$$q(\mathbf{x}) = \frac{1}{2} \mathbf{x}^T \mathbf{Q} \mathbf{x} + \mathbf{c}^T \mathbf{x} + \gamma.$$

Take

$$\mathbf{Q} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \succeq 0.$$

If $\mathbf{c} = \mathbf{0}$,

$$q = \frac{1}{2} x_1^2,$$

so the line $x_1 = 0$ minimizes q .

If $\mathbf{c} = (0, 1)^T$,

$$q = \frac{1}{2} x_1^2 + x_2,$$

so q is unbounded below.

Positive semidefiniteness alone does not decide existence.

$$\mathbf{Q} \succeq 0 \Rightarrow ?$$

When $\mathbf{Q} \succeq 0$, the quadratic q is convex. By Lecture 4,

$$\mathbf{x}^* \text{ is a global minimizer} \iff \nabla q(\mathbf{x}^*) = \mathbf{Q}\mathbf{x}^* + \mathbf{c} = 0.$$

Hence a global minimum is attained exactly when

$$\mathbf{Q}\mathbf{x} = -\mathbf{c}$$

is solvable, or equivalently when

$$\mathbf{c} \in \text{Range}(\mathbf{Q}).$$

Equivalently,

$$\mathbf{z} \in \text{Null}(\mathbf{Q}) = \text{Range}(\mathbf{Q})^\perp \implies \mathbf{c}^T \mathbf{z} = 0.$$

$$\mathbf{Q} \succeq 0, \mathbf{c} \notin \text{Range}(\mathbf{Q}) \Rightarrow \inf q = -\infty$$

Suppose $\mathbf{Q} \succeq 0$ but $\mathbf{c} \notin \text{Range}(\mathbf{Q})$. Then some $\mathbf{z} \in \text{Null}(\mathbf{Q})$ satisfies $\mathbf{c}^T \mathbf{z} \neq 0$. Along $\mathbf{x} = t\mathbf{z}$,

$$q(t\mathbf{z}) = \frac{t^2}{2} \mathbf{z}^T \mathbf{Q} \mathbf{z} + t \mathbf{c}^T \mathbf{z} + \gamma = t \mathbf{c}^T \mathbf{z} + \gamma.$$

Choose the sign of t so that

$$t \mathbf{c}^T \mathbf{z} \rightarrow -\infty.$$

Thus a flat quadratic direction plus a nonzero linear tilt rules out a minimizer.

$$\mathbf{Q} \succeq 0, \mathbf{c} \in \text{Range}(\mathbf{Q}) \Rightarrow \arg \min q \neq \emptyset$$

Suppose $\mathbf{Q} \succeq 0$ and $\mathbf{c} \in \text{Range}(\mathbf{Q})$. Let $\mathbf{x}^* \in \mathbf{R}^n$ be any solution of $\mathbf{Q}\mathbf{x} + \mathbf{c} = 0$. Then $\mathbf{x}^* = -\mathbf{w} + \mathbf{y}$ for some $\mathbf{y} \in \text{Null}(\mathbf{Q})$.

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Suppose $\mathbf{Q} \succeq 0$ and $\mathbf{c} \in \text{Range}(\mathbf{Q})$. Let $\mathbf{x}^* \in \mathbf{R}^n$ be any solution of $\mathbf{Q}\mathbf{x} + \mathbf{c} = 0$. Then $\mathbf{x}^* = -\mathbf{w} + \mathbf{y}$ for some $\mathbf{y} \in \text{Null}(\mathbf{Q})$.

$$\begin{aligned} q(\mathbf{x}^*) &= \frac{1}{2}(\mathbf{x}^*)^T \mathbf{Q}\mathbf{x}^* + \mathbf{c}^T \mathbf{x}^* + \gamma \\ &= \frac{1}{2}(\mathbf{x}^*)^T (\mathbf{Q}\mathbf{x}^* + \mathbf{c}) + \frac{1}{2}(\mathbf{x}^*)^T \mathbf{c} + \gamma = \gamma + \frac{1}{2}(\mathbf{x}^*)^T \mathbf{c} \\ &= \gamma + \frac{1}{2}(-\mathbf{w} + \mathbf{y})^T \mathbf{c} = \gamma - \frac{1}{2}\mathbf{w}^T \mathbf{c}. \end{aligned}$$

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Therefore

$$\min q = \gamma - \frac{1}{2}\mathbf{w}^T \mathbf{c}, \quad \arg \min q = -\mathbf{w} + \text{Null}(\mathbf{Q}).$$

A singular PSD matrix produces an affine family of minimizers.

$$\mathbf{Q} \succ 0 \Rightarrow \arg \min q = \{-\mathbf{Q}^{-1}\mathbf{c}\}$$

If $\mathbf{Q} \succ 0$, then q is strictly convex, $\text{Null}(\mathbf{Q}) = \{\mathbf{0}\}$ and $\mathbf{Q}$ is invertible.

$$\mathbf{x}^* = -\mathbf{Q}^{-1}\mathbf{c}$$

is the unique global minimizer. Equivalently,

$$q(\mathbf{x}) - q(\mathbf{x}^*) = \frac{1}{2}(\mathbf{x} - \mathbf{x}^*)^T \mathbf{Q}(\mathbf{x} - \mathbf{x}^*) > 0 \quad (\mathbf{x} \neq \mathbf{x}^*).$$

In computation, solve $\mathbf{Q}\mathbf{x}^* = -\mathbf{c}$ rather than forming $\mathbf{Q}^{-1}$.

Example: a positive-definite quadratic

Consider

$$q(x_1, x_2) = x_1^2 + x_1x_2 + 2x_2^2 - 2x_1 - 3x_2.$$

Here

$$\mathbf{Q} = \begin{bmatrix} 2 & 1 \\ 1 & 4 \end{bmatrix} \succ 0, \quad \mathbf{c} = \begin{bmatrix} -2 \\ -3 \end{bmatrix}.$$

Solving $\mathbf{Q}\mathbf{x}^* = -\mathbf{c}$ gives

$$\mathbf{x}^* = \begin{bmatrix} 5/7 \\ 4/7 \end{bmatrix}.$$

This point is the unique global minimizer and $q(\mathbf{x}^*) = -11/7$.

Example: the same singular matrix, two outcomes

Let

$$\mathbf{Q} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, \quad \text{Range}(\mathbf{Q}) = \text{span}\{(1, 1)^T\}.$$

For $\mathbf{c} = (-2, -2)^T$,

$$\arg \min q = \{\mathbf{x} : x_1 + x_2 = 2\}.$$

Hence a minimum exists.

For $\mathbf{c} = (-2, -1)^T$,

$$\mathbf{c} \notin \text{Range}(\mathbf{Q}), \quad \inf q = -\infty.$$

Hence no minimum exists.

The linear term decides what happens along $\text{Null}(\mathbf{Q})$.

A local quadratic model suggests a numerical step

Near a current point $\mathbf{x}$, approximate $f(\mathbf{x} + \mathbf{d})$ by

$$m_{\mathbf{x}}(\mathbf{d}) = f(\mathbf{x}) + \nabla f(\mathbf{x})^T \mathbf{d} + \frac{1}{2} \mathbf{d}^T \mathbf{H}_f(\mathbf{x}) \mathbf{d}.$$

If $\mathbf{H}_f(\mathbf{x}) \succ 0$, the model has the unique minimizer

$$\mathbf{H}_f(\mathbf{x}) \mathbf{d}_N = -\nabla f(\mathbf{x}), \quad \mathbf{d}_N = -\mathbf{H}_f(\mathbf{x})^{-1} \nabla f(\mathbf{x}).$$

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In one dimension this becomes

$$d_N = -\frac{f'(x)}{f''(x)}.$$

Numerically, solve the linear system; do not form the inverse explicitly.
This is the Newton step.

Take-home messages

- Stationarity + a PD Hessian $\Rightarrow$ a strict local minimum.
- The certificate is sufficient, not necessary, and says nothing global without extra structure.
- A singular semidefinite Hessian leaves higher-order terms and curved paths to decide.
- Every unconstrained quadratic is classified by the curvature of $\mathbf{Q}$ and whether $\mathbf{c} \in \text{Range}(\mathbf{Q})$.
- Topic 3: numerical methods for unconstrained optimization.

Reference: full proof of second-order sufficiency

Let $m = \lambda_{\min}(\mathbf{H}_f(\mathbf{x}^*)) > 0$. By continuity, choose $r > 0$ such that $B(\mathbf{x}^*, r) \subseteq \mathcal{D}$ and

$$\|\mathbf{H}_f(\mathbf{x}) - \mathbf{H}_f(\mathbf{x}^*)\|_2 \leq \frac{m}{2} \quad \text{on } B(\mathbf{x}^*, r).$$

Hence $\mathbf{d}^T \mathbf{H}_f(\mathbf{x}) \mathbf{d} \geq \frac{m}{2} \|\mathbf{d}\|^2$ there. For $0 < \|\mathbf{d}\| < r$, Taylor's integral formula and stationarity give

$$\begin{aligned} f(\mathbf{x}^* + \mathbf{d}) - f(\mathbf{x}^*) &= \int_0^1 (1-t) \mathbf{d}^T \mathbf{H}_f(\mathbf{x}^* + t\mathbf{d}) \mathbf{d} \, dt \\ &\geq \frac{m}{4} \|\mathbf{d}\|^2 > 0. \end{aligned}$$

Thus $\mathbf{x}^*$ is a strict local minimizer. Apply this result to $-f$ for the maximum case.

Reference: why range and nullspace are orthogonal

Let $\mathbf{Q} = \mathbf{Q}^T$. If $\mathbf{y} \in \text{Range}(\mathbf{Q})$ and $\mathbf{z} \in \text{Null}(\mathbf{Q})$, write $\mathbf{y} = \mathbf{Q}\mathbf{x}$. Then

$$\mathbf{y}^T \mathbf{z} = \mathbf{x}^T \mathbf{Q}^T \mathbf{z} = \mathbf{x}^T \mathbf{Q} \mathbf{z} = 0.$$

Therefore

$$\text{Range}(\mathbf{Q}) \subseteq \text{Null}(\mathbf{Q})^\perp.$$

Rank-nullity gives

$$\dim \text{Range}(\mathbf{Q}) = n - \dim \text{Null}(\mathbf{Q}) = \dim \text{Null}(\mathbf{Q})^\perp.$$

The inclusion and equal dimensions imply

$$\text{Range}(\mathbf{Q}) = \text{Null}(\mathbf{Q})^\perp.$$

Reference: proof of quadratic classification

For $q(\mathbf{x}) = \frac{1}{2}\mathbf{x}^T \mathbf{Q} \mathbf{x} + \mathbf{c}^T \mathbf{x} + \gamma$:

- 1 If $\mathbf{d}^T \mathbf{Q} \mathbf{d} < 0$, then the negative t^2 term in $q(t\mathbf{d})$ gives $\inf q = -\infty$.
- 2 If $\mathbf{Q} \succeq 0$ and $\mathbf{c} \notin \text{Range}(\mathbf{Q})$, choose $\mathbf{z} \in \text{Null}(\mathbf{Q})$ with $\mathbf{c}^T \mathbf{z} \neq 0$; the linear function $q(t\mathbf{z})$ is unbounded below in one direction.
- 3 If $\mathbf{c} = \mathbf{Q}\mathbf{w}$, completing the square gives the minimum and $-\mathbf{w} + \text{Null}(\mathbf{Q})$.

If two vectors $\mathbf{w}_1, \mathbf{w}_2$ solve $\mathbf{Q}\mathbf{w} = \mathbf{c}$, their difference lies in $\text{Null}(\mathbf{Q})$; hence the affine minimizer set and minimum value do not depend on which solution is chosen.